ECONSTOR
Make Your Publications Visible.

A Service of

ZBW Leibniz-Informationszentrum Wirtschaft
Leibniz Information Centre for Economics

Olivares Ugarte, Jazmin Estefania; Bengtsson, Lars

Article

Central characteristics and critical success factors of design thinking for product development in industrial SMEs: A bibliometric analysis

Businesses

Provided in Cooperation with:
MDPI – Multidisciplinary Digital Publishing Institute, Basel

Suggested Citation: Olivares Ugarte, Jazmin Estefania; Bengtsson, Lars (2024) : Central characteristics and critical success factors of design thinking for product development in industrial SMEs: A bibliometric analysis, Businesses, ISSN 2673-7116, MDPI, Basel, Vol. 4, Iss. 4, pp. 843-864, https://doi.org/10.3390/businesses4040046

This Version is available at:
https://hdl.handle.net/10419/338765

Standard-Nutzungsbedingungen:

Die Dokumente auf EconStor dürfen zu eigenen wissenschaftlichen Zwecken und zum Privatgebrauch gespeichert und kopiert werden.

Sie dürfen die Dokumente nicht für öffentliche oder kommerzielle Zwecke vervielfältigen, öffentlich ausstellen, öffentlich zugänglich machen, vertreiben oder anderweitig nutzen.

Sofern die Verfasser die Dokumente unter Open-Content-Lizenzen (insbesondere CC-Lizenzen) zur Verfügung gestellt haben sollten, gelten abweichend von diesen Nutzungsbedingungen die in der dort genannten Lizenz gewährten Nutzungsrechte.

Terms of use:

Documents in EconStor may be saved and copied for your personal and scholarly purposes.

You are not to copy documents for public or commercial purposes, to exhibit the documents publicly, to make them publicly available on the internet, or to distribute or otherwise use the documents in public.

If the documents have been made available under an Open Content Licence (especially Creative Commons Licences), you may exercise further usage rights as specified in the indicated licence.



https://creativecommons.org/licenses/by/4.0/

WWW.ECONSTOR.EU

Mitglied der Leibniz-Gemeinschaft

businesses

MDPI

Review

Central Characteristics and Critical Success Factors of Design Thinking for Product Development in Industrial SMEs—A Bibliometric Analysis

Jazmin Estefania Olivares Ugarte [1,2] and Lars Bengtsson [3,*]

[1] CIRCLE-Centre for Innovation Research, Lund University, P.O. Box 118, 221 00 Lund, Sweden; jazmin_estefania.olivares_ugarte@design.lth.se
[2] Unit of Technology Transfer, Research Institute of the Faculty of Sciences and Technology, Universidad Mayor de San Simon, Cochabamba 2500, Bolivia
[3] Division of Innovation, Department of Design Sciences, Faculty of Engineering LTH, Lund University, P.O. Box 118, 221 00 Lund, Sweden
* Correspondence: lars.bengtsson@design.lth.se; Tel.: +46-708-129-945

Abstract: Design thinking is an innovative methodology that may be applied by small and medium enterprises which emphasizes a human-centered approach to problem-solving, facilitating the creation of novel solutions for complex and open-ended challenges like the development of new products. Despite its recognized importance in business and academia, many industrial SMEs struggle with applying design thinking in their product development processes. This research aims to identify the central characteristics of design thinking and its critical success factors to enable effective application by industrial SMEs. Based on a systematic search in the Scopus database (last searched 14 May 2024) for the relevant literature, which resulted in a selection of 30 published papers, with titles and/or abstract fields containing words of design thinking and small- and medium-sized enterprises in industrial sectors, and a bibliometric analysis of co-word occurrence using VOS Viewer (1.6.20 version), we construct a framework that identifies central characteristics, design thinking principles, criteria, phases, and tools, as well as four dimensions of critical success factors: strategy, culture, competences, and implementation. This finding is new as it applies to industrial SMEs, as compared to extant research's more general orientation. The framework is presented in visual form to facilitate use in developmental workshops with SMEs and supporting actors. This paper ends with notes on limitations and proposals for further research.

Keywords: design thinking; innovation; product development; small and medium-sized enterprises; critical success factors; visual framework

check for updates

Citation: Olivares Ugarte, J.E.; Bengtsson, L. Central Characteristics and Critical Success Factors of Design Thinking for Product Development in Industrial SMEs—A Bibliometric Analysis. *Businesses* **2024**, *4*, 843–864. https://doi.org/10.3390/businesses4040046

Academic Editor: Lincoln C. Wood

Received: 26 August 2024
Revised: 7 November 2024
Accepted: 19 November 2024
Published: 2 December 2024

Copyright: © 2024 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (https://creativecommons.org/licenses/by/4.0/).

1. Introduction

The industrial panorama increasingly calls for multidisciplinary design professionals who can apply design thinking and engineering knowledge in developing new products and services. Design thinking improves an organization's ability to innovate [1,2]. Innovation involves design exploration that results in new products and services, including the creative redesign of existing products, which adds value for the company and the end user. Innovation is recognized as being critical to small- and medium-sized enterprises' (SMEs') survival and development [3]. Despite their relatively weak financial power and limited resources, SMEs may thrive through a commitment to innovation in services and products [4].

The fast evolution of technology has had the effect of dramatically shortening the life cycle of innovative products [5]. As a result, the need has arisen for adopting holistic strategies to maintain competitiveness and offer a more sustainable future [5]. Application of design thinking in enterprises enables them to identify key actors and the users of service [6] to conceptualize, prototype, develop solutions, and enhance communications [7].

Moreover, the extant research on product development processes in industrial SMEs describes them as predominantly being simple, informal, and generally lacking external involvement such as users and customers [8], and, if structured, as following a simplified version of the stage-gate process [9], i.e., idea creation, business cases, and product development and production. The extant research highlights the need for a more structured but also adaptive and easy-to-use approach to product development in industrial SMEs [8].

The extant research suggests that large-sized organizations support and utilize design thinking, but a literature gap exists on how SMEs can successfully adopt design thinking within their long-term strategic management plans [10,11]. While large organizations continue to implement design thinking, SME leaders struggle to adopt design-thinking processes effectively [12,13]. This knowledge gap regarding the particularities for the application of design thinking for product and technology development in SMEs motivates the research presented in this paper, which proposes to build a bibliometric-based framework of the current research to better understand the prerequisites of applying design thinking for product development in industrial SMEs.

The question guiding this research is the following: Which are the central characteristics and critical success factors that are needed to facilitate the effective application of design thinking for product development in industrial SMEs?

The aim is to construct a framework for the application of design thinking in industrial SMEs that can provide guidance to industrial SME managers and other stakeholders, supporting the development of SMEs, i.e., giving an overview of the current research as well as identifying the most salient issues in the application of design thinking for product development.

2. Conceptual Background

Definition of Design Thinking

Design thinking is here defined as an approach that supports solving various problems by harmonizing user desirability, economic viability, and technological feasibility [14–16]. Design thinking combines "empathy for the context of a problem, creativity in the generation of insights and solutions, and rationality in analyzing and fitting various solutions to the problem context" [17] by inviting the end user/consumer to be a part of the innovation process [18].

Design thinking is emerging in the management literature as a concept that promises innovation through a more user-centered approach that suggests that companies can learn from the way designers think and work [14,19]. As a result, it has been introduced in many different organizational settings, such as SMEs [20], to solve complex and open-ended problems like new product development. According to [21], there is a growing interest in design thinking (DT) among managers because the integration of the DT process into an SME's product development strategy will improve its competitive position [22]. However, the integration of DT into the product development process can be approached in various ways. To fully comprehend the potential benefits of DT for product development, it is essential to understand the different manifestations of DT. Scholars have identified three primary forms of DT application: as a mindset, as a process, and as a toolbox [23,24].

When considered as a mindset, DT is characterized by several key principles, including a strong focus on both explicit and latent customer and user needs, as well as an emphasis on prototyping [24]. Nonetheless, it has been argued that applying these principles in isolation—without a structured framework—can be overly challenging for novices such as inexperienced SME managers [24]. Consequently, a structured process is necessary to facilitate novice comprehension of what DT is and how it can contribute to the product development process.

Additionally, various collections of design tools exist, catering to both practitioners [25] and academics [26]. The deployment of appropriate methods is a critical success factor in DT projects [24]. Therefore, it is imperative that product development teams possess a thorough understanding of how to apply these methods effectively. Thus, the generation

of a research-based framework with the central characteristics and critical success factors of DT may facilitate the SME managers' understanding of how it works and how it can be applied successfully. This is especially important for SMEs with limited financial and other resources. While larger companies usually can withstand the consequences of failed product development projects, SMEs have a much lower-level resilience against such failures. Thus, learning lessons from previous product development projects are essential to guide SMEs comprehensively from the application of design thinking into the product development process.

DT is distinct from traditional product development methodologies due to its greater emphasis on comprehensively understanding the user. This approach employs methods such as user profiles, journey maps, and co-design to gain deeper insights into users' needs and experiences. DT aims to reframe problems by identifying the fundamental issues and underlying reasons behind user preferences and behaviors. It emphasizes generating a variety of solutions, utilizing rapid prototyping, and adopting a strategy of "failing early" to achieve success more swiftly [27]. This last strategy allows SMEs not to invest a lot of financial resources until the prototype is approved by the end users to start the product development. Typically, the DT process stops when you have a fully functional and user-approved prototype.

The successful application of DT can be achieved by identifying its critical success factors (CSFs). These are defined as items or actions that should be present in a particular project or situation to be successful [28]. By identifying the CSFs, companies can proactively implement necessary measures to avoid potential failures or problematic areas, thereby enhancing the success rate of implementing DT. The research of de Paula et al., [29] contributes to the identification of 20 CSFs divided into four key dimensions to understand how to successfully implement design thinking in business organizations.

Table 1 shows more details about the theoretical origin of 20 CSFs grouped into four dimensions, namely, strategy, culture, implementation, and competences.

Table 1. Critical success factors in DT application. Source: [29] de Paula et al. (2019).

Dimensions	Factors	Source
Strategy	Secure management support	[26,30–32] Carlgren et al. (2016), Holloway (2009), Rosensweig (2011), Hassi and Laakso (2011)
	Having fundamental guidelines for design thinking	[26,31,33] Rosensweig (2011), Kimbell (2009), Hassi and Laakso (2011)
	Ensure fundings for design thinking initiatives	[14,32] Carlgren et al. (2016), Brown (2008)
	Having clear metrics	[21,34] Carlgren, Rauth, et al. (2016), Liedtka (2011)
Culture	Diversity orientation	[26,30,31] Holloway (2009), Rosensweig (2011), Hassi and Laakso (2011)
	Foster empathy	[14,33–35] Brown (2008), Kimbell (2009), Liedtka (2011), Lockwood (2009)
	Ability to handle ambiguous situations	[36–38] Dunne et al. (2006), Gloppen (2009), Sato et al. (2010)
	Ability to handle complexity and uncertainty	[34,39–41] Liedtka (2011), Boland and Collopy (2004), Cooper et al. (2009), Dew (2007)
	Establish collaboration and cross-functional teams	[36–38] Dunne et al. (2006), Gloppen (2009), Sato et al. (2010)
Implementation	Provide necessary material	[42,43] Micheli et al. (2012), Carlgren et al. (2014)
	Innovation spaces	[14,43] Carlgren et al. (2014), Brown (2008)
	Establish flexible and responsive processes	[24,30,31] Holloway (2009), Rosensweig (2011), Brenner et al. (2016)
	Integrate DT into NPD and related processes	[31,42] Micheli et al. (2012), Rosensweig (2011)
	Apply lessons learned from past projects	[23,24,32] Wölbling et al. (2012), Carlgren, Elmquist, et al. (2016), Brenner et al. (2016)
	Access to the user	[14,30,44] Brown (2008), Holloway (2009), Ward et al. (2009)

Table 1. *Cont.*

Dimensions	Factors	Source
Competences	Provide training on DT	[31,32,42] Micheli et al. (2012), Carlgren, Elmquist, et al. (2016), Rosensweig (2011)
	Collaborative initiative with key partners	[31,42] Micheli et al. (2012), Rosensweig (2011)
	Create DT awareness	[36,38,43] Dunne et al. (2006), Sato et al. (2010), Carlgren et al. (2014)
	Enable the optimal team skills	[14,32,45] Carlgren, Elmquist, et al. (2016), Brown (2008), Seidel and Fixson (2013)
	Include DT principles into everyday work	[14,32,34] Carlgren, Elmquist, et al. (2016), Brown (2008), Liedtka (2011)

3. Research Design and Methodology

This study adhered to the PRISMA 2020 guidelines in its relevant parts. This paper applied a two-step methodology: a systematic search of the relevant literature using a search string in the database Scopus (last search performed on 14 May 2024) and then a bibliometric analysis of keywords and abstracts in the selected papers. In doing so, the characteristics and variables involved in the process of DT application for the development of new products/technology for SMEs were identified. Bibliometric analysis is an effective method of summarizing and synthesizing the literature [46]. The process of systematic search was divided in four steps: (1) define the search string, (2) filter the search outcomes according to the selection criteria, (3) exclude the papers that do not follow the string domain and include additional papers according to snowball sampling, and (4) analyse the included literature to gain insights.

3.1. Definition of Search String

The search query was composed of the following keywords: "design thinking" * AND "innovation" AND (sme* OR smes* OR "small and medium-sized enterprise" OR "small and medium enterprise" OR "small medium enterprise" OR "small and medium-sized firm" OR "small and medium firm" OR "small firm" OR "medium firm"*).

3.2. Filtering Criteria and Process

The search string included articles, conference papers, and book chapters in engineering, business, management and accounting subject areas in English and German.

3.3. Exclusion of Papers and Inclusion of Additional Papers

The excluded papers were studies related to digitalization, arts and humanities, and tourism and education, i.e., services not covered in this research. Hence, the focus was on industrial SMEs. The search query yielded 25 articles, of which 17 articles were selected because they showed factors and prerequisites that facilitated the application of DT for the development of products. Additionally, 13 articles were collected by snowball sampling, making a total of 30 articles that were published between 2011 and 2023. This interval of years was obtained based on the search string filter, i.e., the first paper that appeared and met the inclusion criteria was published in 2011 and the last paper was published in 2023. Figure 1 summarizes the research method followed for this research. The 30 articles filtered from Scopus and the snowball sampling are listed in Table 1 in Appendix A. The datasets included in the systematic search of the literature review are small, but we applied a rigorous exclusion and inclusion process that guaranteed the quality and relevance of the literature reviewed [47].

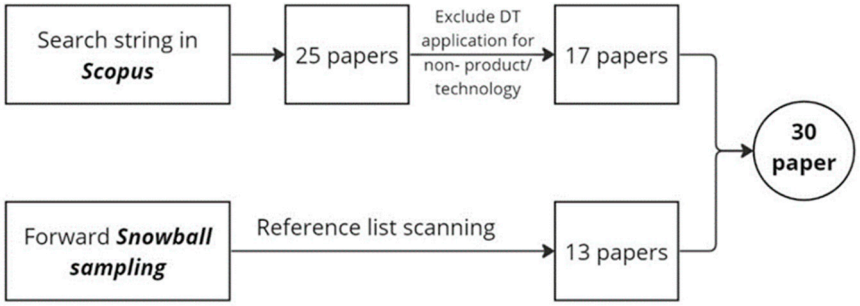


Figure 1. Selection process for the papers included in the study.

3.4. Analysis of Eligibility

The consistency of the information that matched the main criteria followed these main conditions:

- The title, abstracts, and keywords of the literature must include the terms 'design thinking' AND 'SMEs';
- The application of the use of design thinking should be focused on developing new products or new technologies.

The first author independently performed the searches in the database and performed a first selection of included and excluded papers. The second author checked and confirmed the selection of the papers. The first author collected the data from the selected papers, the title, and the abstract fields and prepared the data for analysis.

4. Results

The bibliometric analysis of keywords and abstracts aimed to identify and describe the relevant prerequisites needed to facilitate the application of DT in the context of industrial SMEs.

4.1. Bibliometric Analysis

The central characteristics and critical success factors were identified based on the visualization of the group network obtained in the key word co-occurrence analysis of the titles and abstract fields of the selected papers, as shown in Figure 2. The potential contributions of content co-occurrence analyses included the syntheses of the literature based on themes and related publications that form disparate research streams that are mapped into a taxonomy [48]. The key word co-occurrence analysis of titles and abstract fields of the 30 papers were filtered and uploaded in Research Rabbit, which is a 'citation-based literature mapping tool', to obtain the database as a Research Information Systems Document (.ris). No data were missing; all papers had titles and abstract fields. This database was exported using VOS viewer software (1.6.20 version, CWTS BV company), a tool for constructing and visualizing bibliometric networks. Bibliometric analysis allows us to model the evolution of concepts and identify the topics studied in a field of inquiry [49,50]. A bibliometric network consists of nodes and edges. The nodes can be publications, journals, researchers, and keywords, and the edges indicate relation between pairs of nodes. VOS viewers used techniques for mapping and clustering the nodes in a network [51]. A cluster is a set of closely related nodes.

The network of the keyword co-occurrence analysis (co-word analysis) enables us to visualize the relationships among keywords or topics [49–51]. Co-word analysis is a text-mining technique that analyzes the 'co-occurrence' of pairs of keywords in the selected documents [51]. This analysis assumes that keywords that frequently appear together (i.e., co-occur) in the same document bear a relationship to one another.

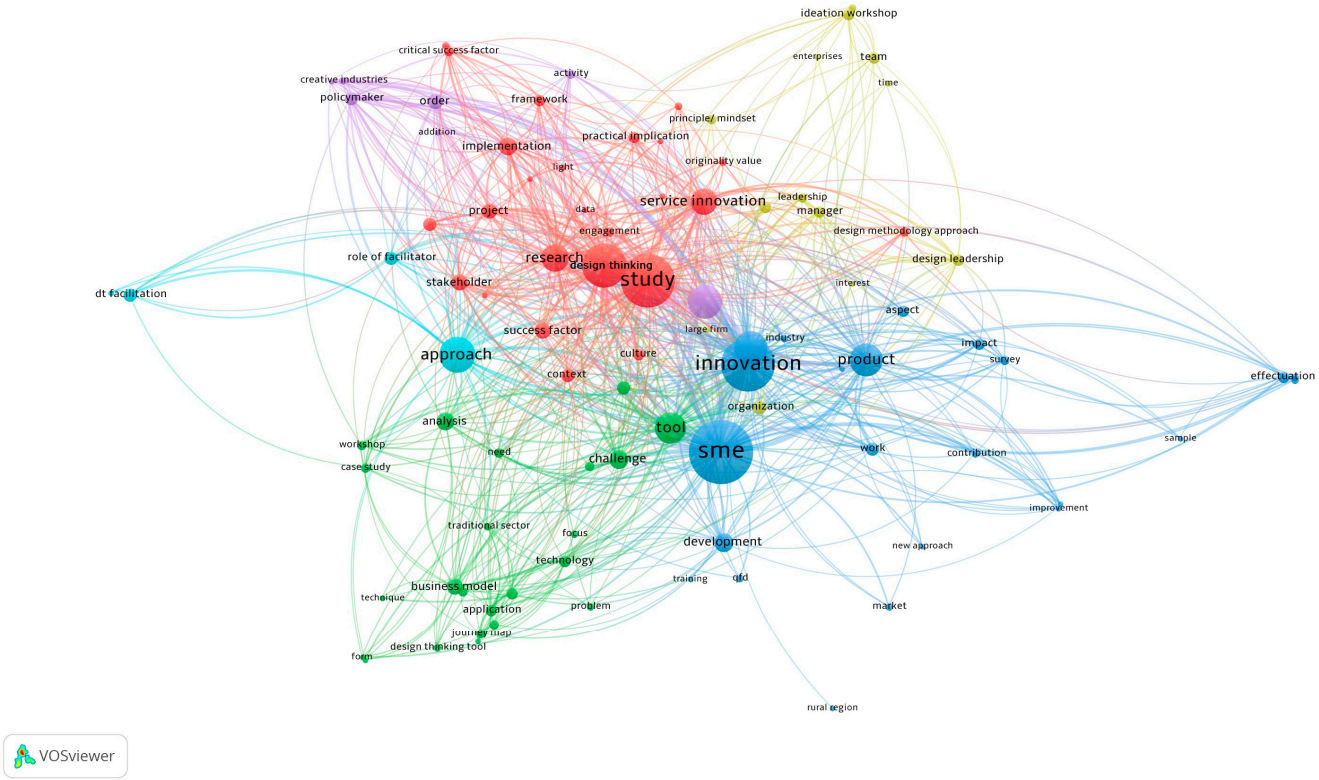


Figure 2. Visualization of DT characteristics based on the co-occurrence analysis of abstracts and keywords. Source: prepared by the authors based on Research Rabbit data and VOS viewer (version 1.6.20 software of CWTS BV company owned by Leiden University).

The parameters for the software are the minimum number of key word co-occurrences defined as 2, as recommended by [52] Baier–Fuentes et al. (2018), with a minimum threshold of 95 items and six clusters. The co-word analysis indicates the key terms most frequently used in literature (bigger nodes in Figure 2), like design thinking, innovation, SME, study, approach, process, product, and tool.

The six clusters identified in Figure 2 can be described as follows, in order of the highest to lowest degree of absolute frequency:

The red cluster with 25 items (26.32% of the total items) focuses on the dimensions of critical success factors like culture and the implementation of design thinking. Additionally, it is recognized that some critical factors like management support inside the strategy dimension and the knowledge exchange among stakeholders as part of the culture dimension.

The green cluster with 23 items (24.21%) is related to the application of design thinking tools for different levels of innovation, like the application/adoption of technology, business models, and ecosystems. The tools identified are journey map and point of view etc.

The blue cluster with 23 items (24.21%) relates to both design thinking tools and critical success factors, with an emphasis on SMEs, such as QFD and surveys applied for the development of products for SMEs. The critical success factors identified belong to the competence dimension, like training and development.

The yellow cluster with 13 items (13.68%) focuses on principles/mindsets, such as experimentation, team organization, and design leadership to improve the product development of enterprises.

The purple cluster with seven items (7.37%) emphasizes the process of activating new avenues of innovation, like design thinking as applied to inexperienced actors in the design field. Design thinking can be learned and adopted by SMEs from large firms and by policymakers to enable SMEs to co-create with creative industries [53].

Finally, the light blue cluster with four items (4.21%) shows the emphasis on design thinking facilitation criteria like desire, feasibility, viability and sustainable/circular ap-

proach. Additionally, it highlights the role of the facilitator and the taxonomy of the design thinking approach.

In summary, the central characteristics of design thinking identified by the six clusters in the co-word analysis are as follows: tools, levels of innovation, principles/mindsets, phases or processes of design thinking, and facilitation criteria. In addition, there is the cluster of critical success factors.

4.2. Details of Central Characteristics and Critical Success Factors for the Application of Design Thinking Identified Based on Co-Word Analysis

Table 1, found in Appendix A, highlights the themes (clusters) of central characteristics and critical success factors for the integration of DT in SMEs as identified in the bibliometric analysis of keywords in the 30 papers, i.e., principles/mindsets, facilitation criteria, innovation phases, levels of innovation, tools, and CSFs. Table 1 provides a novel view for DT application in SMEs based on the selected papers. Figure 3 shows the absolute frequencies for the central characteristics of design thinking obtained in the analysis of the 30 papers detailed in Table 1.

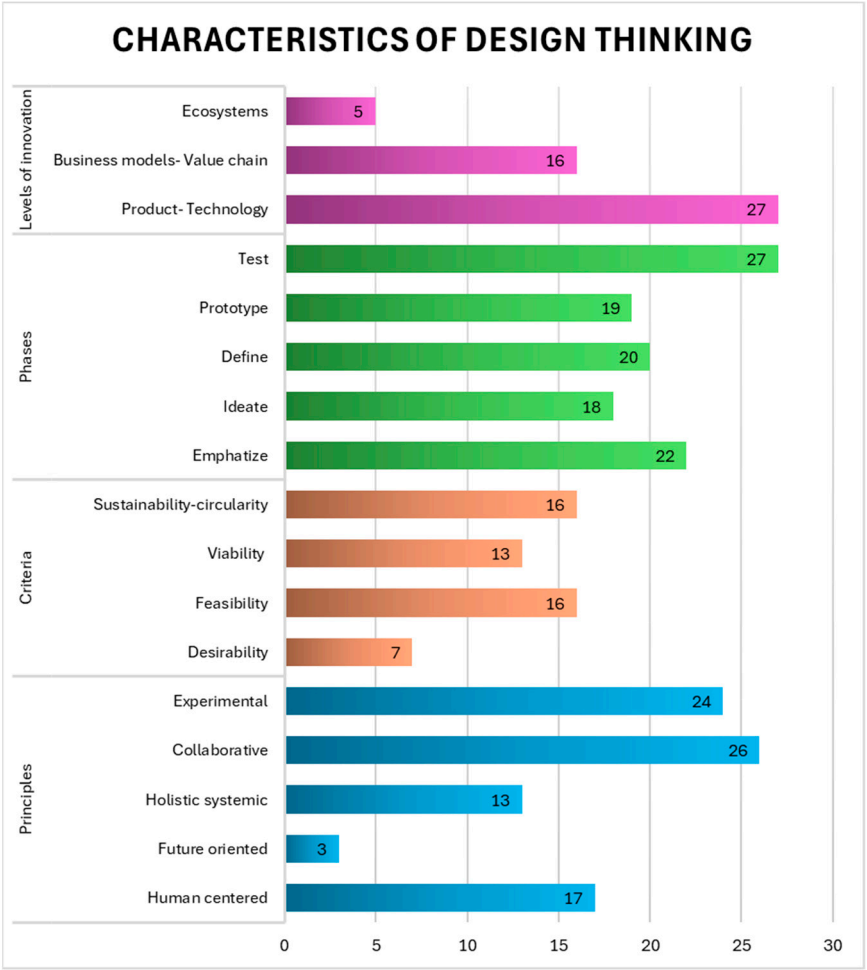


Figure 3. Absolute frequencies for central characteristics of design thinking.

The *level of innovation* relates to the product/technology, business model, or ecosystem levels [54,55]. The analysis on the *level of innovation* shows in the purple bars of Figure 3 that at least 90% of all the selected papers used design thinking for developing a product or some technology. While designers have traditionally focused on technology and product design, the intersection of products, services, and business models is emerging as a significant area of emphasis in both business and design research and practice [56]. This shift is attributable to the role that business models play in determining a product's impact on

customer success [57] and its environmental footprint [58]. Moreover, design is increasingly recognized for its potential to facilitate the transition of broader ecosystems toward circular cities, achieved through the redesign of products and business models, and the application of design thinking to implement innovative projects [59].

The *principles/mindsets* of design thinking are human-centered, future-oriented, holistic, systemic, collaborative and experimental [21]. Collaborative, experimental, and human-centered principles/mindsets are often recognized as part of the culture of design thinking for the development of new products [21]. Holistic, systemic, and future-oriented principles are recognized as belonging to design thinking mentality [26] and are less frequently mentioned in this analysis. All these principles are recognized in the critical success factors identified by [29] of the culture dimension, including the ability to handle ambiguous situations (experimental), the ability to establish collaboration and cross-functional teams (collaborative), diversity orientation (holistic/systemic), the ability to handle complexity and uncertainty (future-oriented) and the ability to foster empathy (human-centered). Additionally, the collaborative principle is the most recognized in the papers, as shown in the blue bar of Figure 3, because the design process is often co-creative involving frequent interactions with multiple customers and stakeholders [60].

The *criteria* are focused on desirability, feasibility, viability, and sustainability/circularity. The desirability criterion refers to what people need and/or want; feasibility refers to what is doable from a technical, technological, and/or operational standpoint; and viability refers to what is possible financially and/or economically for the innovating organization [61]. These three criteria are central to design thinking and relevant to circular and sustainable innovation as well [62]. The criteria of feasibility and sustainability/circularity, as shown in the orange bars of Figure 3, are predominant for building new ideas to pass from linear production to circular economy [63]. The traditional design-thinking tools predominantly build upon the feasibility and desirability of new ideas and less on viability (except the business model canvas), or circularity and sustainability, whereas the circular economy-specific tools build upon the feasibility, viability, and circularity criteria [63].

The *Innovation phases* are summarized into five steps: empathize, define, ideate, prototype, and test, based on HPI at Stanford (d. school) [64]. This model of design thinking has gained the most attention [65] because of its roots in academia [66,67], which facilitates the learning processes of diverse stakeholders like SMEs and large firms for DT implementation. With empathy, designers understand users and their actions [68]. In the definition phase, the collected information is processed, and the challenge is defined [69]. In the ideation phase, rough ideas are developed, while in the prototyping phase, a functional model that helps to verify the design is created [68,69]. The last phase is testing in real conditions, which can be carried out at all stages of the process, and the purpose is to obtain feedback based on the prototype [69].

Regarding the 20 *tools* identified in Table 1, most of them (49%), are used for the *test* and *empathize* phases, as shown in the green bars of Figure 3. The tools recognized in all phases are the quality tools (virtual reality), the customer side, the supplier side, participant observation, interviewing, co-creation, experimentation, focus groups, and building a point of view. Virtual reality tools refer to the application of computer-based digital techniques to simulate the sensation of "transferring" users—or more accurately, their consciousnesses—to a location different from their actual physical presence. To achieve this, users are immersed in a virtual environment (VE) constructed through three-dimensional computer graphics, which allows them to interact with various elements within this simulated space [70]. Design thinking tools, such as contextual interviews and co-creation with diverse stakeholders, facilitate the development of effective strategies and the identification of business opportunities. These tools also support the application of rapid prototyping to concretize organizational visions and strategies [10].

The tool used in the *empathize*, *define*, *prototype*, and *test* phases are customer journey maps, visualization tools that provide the means and opportunity for mapping and understanding the multidimensional experiences of a customer when interacting with a business

or a product [71]. Tools used in the *empathize*, *ideate*, and *define* phases are stakeholder mapping and the stakeholder profiling canvas. In these phases, DT integrates live stakeholder engagement, providing frequent opportunities for the interpretation of viewpoints and for needs to be corrected and re-balanced [72]. Stakeholder mapping is a tool that supports the highlighting of less-emphasized groups to bring alternative viewpoints, perspectives, and areas of need into the horizon of decision makers [7], thus assisting in addressing the question regarding 'whom' to empathize with. Stakeholder profiling facilitates the ability to understand the operating context, environment, and needs of a given party [73].

The tool used in the *empathize*, *ideate*, and *test* phases involves the job to be done (JTBD) approach [74], the objectives of which are providing understanding about the jobs of each persona, understanding what the key and secondary tasks are, and describing how the firm's products/service fit into the jobs of the stakeholder.

The tools used in the *empathize*, *define*, and *test* phases are the empathy map and persona. The persona tool is especially helpful in the early phases of product design [75]. The personas are constantly used to evaluate the ideas and prototypes regarding their needs and pains [76]. Also, the empathy map is used to understand behaviors, decision-making, and pains/gains of the most important stakeholders to build the customer value proposition [77].

In the *ideate*, *prototype*, and *test*, the Quality Function Deployment (QFD), storyboards, user stories, brainstorming and rapid prototyping tools are used. The utilization of the QFD tool enables the early integration of 'manufacturability' considerations into the design process, rather than addressing these issues retrospectively after the development of a prototype [78].

The tools recognized in the *test* phase are the contradiction matrix and the environment mapping tool. The use of the contradiction matrix and environmental mapping tools help SMEs to systematically generate innovations in the business model, to reduce the solution space in a structured way, and to evaluate the resulting solutions according to their advantages and disadvantages [79].

Figure 4 shows an analysis of the degree of co-occurrence of the DT tools, revealing the tools most frequently used for development of new products and technology.



Figure 4. Degree of occurrence of design thinking tools for the development of new products/technology.

The tool most frequently mentioned in the articles is the user/customer journey map, which is predominantly focused on the principles of user-centered, systemic thinking, collaboration, and experimentation with multiple stakeholders and are to a lesser extent future-oriented. The persona, interviewing, and co-creation tools are the second most-used tools for prototype development. The third place is occupied by the job to be done and participant observation tools.

Finally, the *critical success factors* for the effective implementation of design thinking are summarized in the conceptual framework proposed by [29]. Table 1 was the starting point for performing an absolute frequency analysis, as detailed in Figure 5, to show the underlying factors recognized for design thinking implementation for product development.

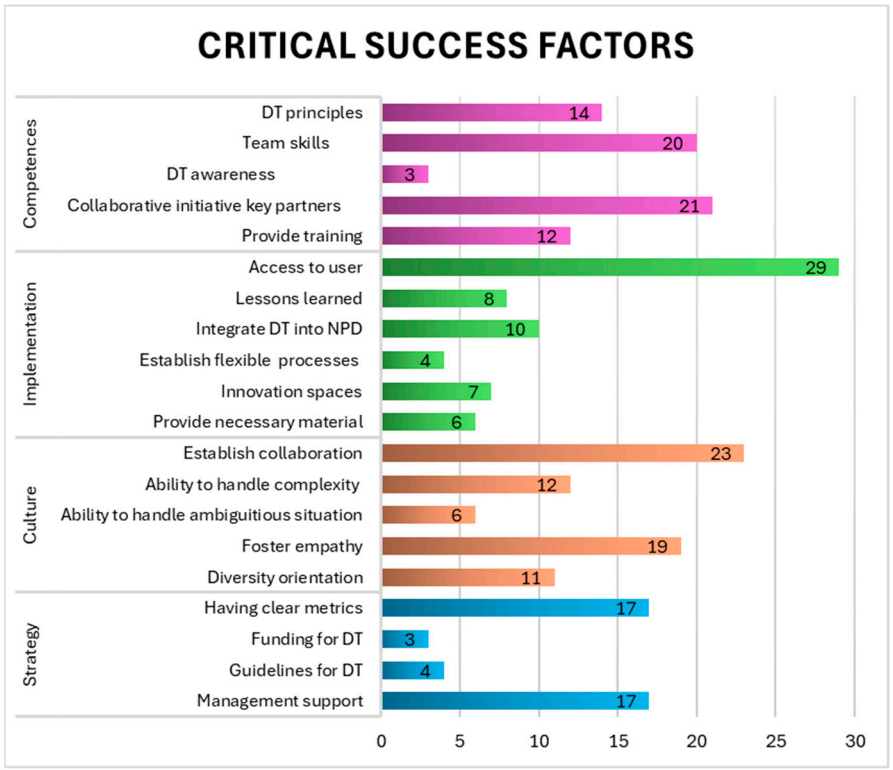


Figure 5. Absolute frequency of the critical success factors.

First, the *strategy* dimension highlights the factors of management support that must be met to the necessary resources to perform DT-related activities [21] and have clear metrics. Second, the *culture* dimension emphasizes the CSFs of establishing collaboration through cross-functional teams to tackle complex and wicked problems by gaining knowledge from many fields and disciplines [37] and by fostering empathy with users. This factor means recognizing and understanding other people's sensations, emotions, thoughts, motives, and personality traits [14,21,80].

Third, the dimension of *implementation* recognizes the factor of access to the user, which is the starting point of DT that consists of the observation and analysis of the situation with a focus on the user [14]. Finally, the *competence* dimension recognizes the factors of collaborative initiative with key partners like universities and/or research institutes to promote projects [29] and team skills factors to bring different perspectives to project development [45].

5. Discussion

The systematic search for relevant papers has shown that the current research regarding the implementation of DT in industrial SMEs is limited, i.e., just 30 academic papers have been published so far. Thus, the research knowledge regarding the application of DT in the specific context of an industrial SME, such as time and resource limitations, is still limited. Building on the bibliometric analysis of the 30 papers, we have identified the central characteristics and CSFs of DT implementation, as summarized in Figure 6. The central characteristics for the implementation of DT in enterprises are divided in principles/mindsets, criteria, phases, and tools. Additionally, CSFs are divided into four dimensions: strategy, culture, implementation, and competences.

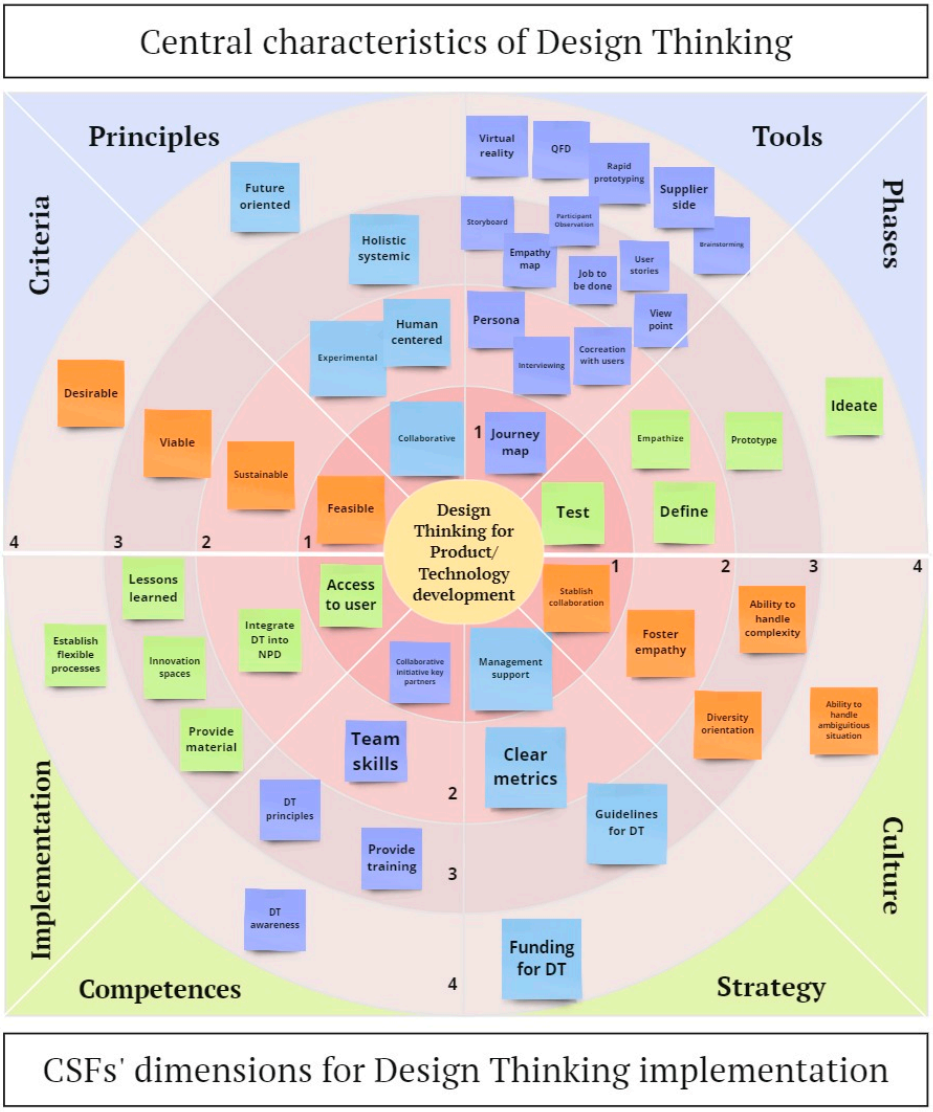


Figure 6. Visual framework of the central Characteristics and Dimensions of critical success factors for the Implementation of design thinking in industrial SMEs based on the co-word analysis. Note. Numbers 1-4 are the levels from most (1) to less (4) frequently used of variables.

Moreover, some of the components in the central characteristics and CSFs occur more frequently than others, indicating their relative importances in an SME context. The key words that are closer to the center in Figure 6 are those that appear most frequently in the selected papers and are applied to the development of new products/technology, in industrial SMEs. While the occurrences of the keywords for characteristics and CSFs are not different from the extant research [23,24], their frequencies indicate an order that is unique to industrial SMEs.

The framework detailed in Figure 6 provides a visual representation of the central characteristics and CSFs of DT for effective application in product development processes in industrial SMEs. While the framework gives an overview of all the components of the characteristics and CSFs that are important for the effective application of DT, some are especially important, or perhaps challenging, in the SME context, as the research mentions these frequently. Compared to the extant research on DT application [23,24], this study shows an order of importance for the characteristics and critical success factors. For instance, the most-mentioned characteristics and CSFs concern the integration of the user in various ways (e.g., user access, foster empathy, testing, journey maps, personas, human-centered) and collaboration with others (e.g., being able to initiate collaboration

with key partners, establishing a culture of collaboration, co-creation with users, feasibility) indicating that these two application dimensions (integrating users, collaboration with others) might prove to be especially challenging for the SME. The benefits of access to users in various ways facilitates a deeper understanding that is well-documented in the literature so that technically oriented individuals and teams may derive novel insights for more concerted and user-oriented product ideas by incorporating this perspective [81]. Successful collaborations require trust and respect for users and other key partners in combination with establishing a collaborative organizational culture as suggested by Björk et al. (2014) [82].

The importance of visualization tools such as customer journey maps, storyboards, and personas support a deeper understanding of stakeholder needs and intentions [83]. The manifestations of tangible thoughts, like sketches, diagrams, and scenarios, inspire and enable communication with the SME's internal users and key partners, as well as with supporting actors such as consultants and universities. Thus, in the same spirit, the research-based framework of the central characteristics and CSFs is a visual framework. The Post-it notes in the visual framework also make it possible to adapt to specific SME contexts and situations, i.e., they may be rearranged in terms of relevance and importance in development workshops.

6. Conclusions

This paper's research question was concerned with identifying the central characteristics and critical success factors that are needed to facilitate the effective application of design thinking for product development in industrial SMEs, with the aim of constructing a framework for the application of DT in industrial SMEs that can provide guidance to industrial SME managers and other stakeholders supporting the development of SMEs, i.e., giving an overview of the current research as well as identifying the most salient issues in the application of DT for product development in industrial SMEs.

After a systematic search for the relevant research literature, we performed a bibliometric analysis of the titles and abstracts of the selected 30 academic papers. The identified central characteristics of DT application are divided into four aspects: principles, criteria, phases, and tools that enable an effective application of industrial SMEs for product development. Furthermore, the identification of CSFs is divided into four dimensions: culture, competences, strategy, and implementation.

The design thinking phases are empathize, ideate, define, prototype, and test. The facilitation criteria include desirability, feasibility, viability and sustainability, and circularity. Finally, the following principles that make design thinking suitable to tackle complex prototype innovation challenges are the qualities of being human-centered, future-oriented, holistic systemic, collaborative, and experimental. In addition, we mapped 20 tools used in DT.

The analysis indicated that the integration of users in various ways and the competence and culture of collaboration are especially important, or perhaps challenging, for SMEs to master. In the spirit of DT, we presented the research-based framework in visual form, as shown in Figure 6 above, in order to be used as a visual tool for SME managers and supporting actors in the process of applying DT in their product development processes.

7. Limitations and Future Research

As discussed above, DT in industrial SMEs is an emerging topic and is in need of further study because of the ambiguity surrounding the way it is implemented, utilized, and discussed in the specific context of SMEs. Our study has some limitations which warrant further research. First, the framework is based on the 30 selected research papers. Our way of identifying the relevant papers, such the database used (Scopus), the search keywords, and the snowballing, could result in relevant papers being missed in the selection of papers. Thus, future bibliometric analysis or systematic reviews could be complemented with other databases and keywords. Second, we have used a bibliometric analysis of titles and abstract

fields to identify central characteristics and CSFs. Even though this is a standard analytical tool, there is always a component of judgement in these types of analyses. Thus, other analytical tools might be used in future reviews to check the robustness of our findings. Third, the research-based framework has been presented in visual form to give a summary and overview of the research field, with the aim of facilitating use in workshops with industrial SME managers and other stakeholders who intend to apply DT or are struggling with its application. However, we have not tested the visual framework in workshops with SMEs or other stakeholders. Thus, future research should test this and other similar frameworks in workshops to validate their values and effects.

Furthermore, the integration of topics like artificial intelligence in design thinking could potentially eliminate tedious processes, improve user-centricity, and stimulate creativity. While designers will remain responsible for interpreting and integrating human emotions, behaviours, and values, AI may augment their ability to develop innovative, personalized, and impactful solutions. Thus, this could be a future topic for analysis in the context of design thinking for product development in industrial SMEs.

Author Contributions: J.E.O.U. assumes the role of the primary author, responsible for crafting the majority of the text. L.B. contributed by assisting in the formulation of the research design, reviewing the introduction, offering insights and writing the results, discussion, and conclusion segments, as well as meticulously reviewing and enhancing the manuscript at each stage of the research's development. Conceptualization, J.E.O.U. and L.B.; methodology, J.E.O.U. and L.B.; software, J.E.O.U.; validation, J.E.O.U. and L.B.; formal analysis, J.E.O.U.; investigation, J.E.O.U. and L.B.; resources, L.B.; data curation, L.B.; writing—original draft preparation, J.E.O.U.; writing—review and editing, L.B.; visualization, J.E.O.U.; supervision, L.B.; project administration, L.B.; funding acquisition, N/A. All authors have read and agreed to the published version of the manuscript.

Funding: The author(s) disclose receipt of the following financial support for the research, authorship, and/or publication of this article. This work was supported by the agreement between Sweden and the Universidad Mayor de San Simon with the Research Cooperation Programme "Strengthening Institutional Research Capacities in Universidad Mayor de San Simon 2021–2025," Swedish International Development Cooperation Agency (SIDA) Contribution 13486. The funding is for contributing to advancement in universal knowledge and developing postgraduate studies in science in order to further scientific research, technological development, and innovation activities.

Data Availability Statement: Some data are unavailable due to privacy or the ethical restriction of permission to publish from the copyright holder for any previously published content.

Acknowledgments: The authors express gratitude to the administrative and technical support for the co-word analysis.

Conflicts of Interest: The authors declare no conflicts of interest. The funders had no role in the design of the study; in the collection, analysis, or interpretation of data; in the writing of the manuscript; or in the decision to publish the results.

Appendix A

Table 1. Central characteristics of design thinking for the development of technology for SMEs.

			Principles					Criteria				Phases					Levels of Innovation			Tools	Critical Factors			
N	Year	References (in Chronological Order)	Human Centered	Future Oriented	Holistic Systemic	Collaborative	Experimental	Desirability	Feasibility	Viability	Sustainability–Circularity	Empathize	Ideate	Define	Prototype	Test	Product–Technology	Business Models–Value Chain	Ecosystems	Design Thinking Tools	Strategy	Culture	Implementation	Competences
1	2023	[84] Gao, B., & Yu, K. (2023). Knowledge exchange in SMEs service innovation with design thinking. Management Decision, 61(7), 2029-2049. https://doi.org/10.1108/MD-06-2022-0795	x			x			x	x	x						x	x		N/A	Management support	Establish collaboration	Access to user	-Collaborative initiative key partners -DT principles
2	2023	[85] Baumkötter, B., Ellenbürger, J., Glemser, M., Raschi, M., Reinschmiedt, L., Schaller, J., & Palm, D. (2023). Neue Produkte erfordern neue Denkweisen zur Einführung: Entwicklung eines neuartigen Markteinführungskonzeptes. Zeitschrift für wirtschaftlichen Fabrikbetrieb, 118(4), 232-236. https://doi.org/10.1515/zwf-2023-1048	x			x	x				x						x			N/A	Management support	Establish collaboration	-Access to user -Integrate DT into NPD	DT principles
3	2023	[79] Rittershaus, P., Renner, M., & Aryan, V. (2023). A conceptual methodology to screen and adopt circular business models in small and medium scale enterprises (SMEs): A case study on child safety seats as a product service system. Journal of Cleaner Production, 390, 136083.			x	x	x				x					x		x		Contradiction matrix	-Guidelines for DT -Clear Metrics	Establish collaboration	Access to user	Collaborative initiative key partners
					x	x	x				x					x		x		Environment mapping tool				
4	2022	[86] Storm, J., & Smith, A. (2022). Empathize with Whom? Adopting a Design Thinking Mind-Set to Stimulate Sustainability Initiatives in Chinese SMEs. Sustainability, 15(1), 252.	x			x	x				x	x	x	x					x	Stakeholder Mapping	Management support	-Diversity orientation -Foster empathy	Access to user	-Collaborative initiative key partners. -Team skills -DT principles
			x			x	x				x	x	x	x					x	Stakeholder Profiling Canvas				
5	2022	[70] Starzyńska, B., Kujawińska, A., Górski, F., & Buń, P. (2022). VIRTUAL QUALITY TOOLBOX AS AN INNOVATIVE SOLUTION SUPPORTING LIFELONG LEARNING. International Journal for Quality Research, 16(4).	x			x	x	x				x	x	x	x	x	x		x	Quality tools like VR (Virtual reality)	Having clear metrics	Foster empathy	-Access to user -Provide necessary material	-Provide training. -Team skills
6	2021	[83] Paay, J., Kuys, B., & Taffe, S. (2021). Innovating product design through university-industry collaboration: Codesigning a bushfire rated skylight. Design Studies, 76, 101031.				x	x		x			x		x	x	x	x			Customer journey maps,	Management support	-Diversity orientation -Foster empathy -Establish collaboration	-Access to user -Provide necessary material -Innovation spaces	-Collaborative initiative key partners. -Team skills
						x	x		x				x		x	x	x			Storyboards				
						x	x		x			x		x		x	x			Persona				
7	2021	[87] Whelan, L., Kiernan, L., Morrissey, K., & Deloughry, N. (2021, May). Measuring the success factors of strategic design implementation. In 2021 IEEE Technology & Engineering Management Conference-Europe (TEMSCON-EUR) (pp. 1-6). IEEE.							x	x							x	x		N/A	Having clear metrics	-Diversity orientation -Establish collaboration	Provide necessary material	-Collaborative initiative key partners. -Team skills
8	2020	[88] Lugnet, J., Ericson, Å., & Wenngren, J. (2020). Innovation supports for small-scale development in rural regions: a create, build, test and learn approach. International Journal of Product Development, 24(1), 30-42.			x	x	x										x			N/A	-Management support -Guidelines for DT	Foster empathy	-Access to user -Provide necessary material -Lessons learnt	-Provide training. -Team skills
9	2020	[53] Melazzini, M., Campodall'Orto, S., Carella, G., Vignati, A., & Zurlo, F. (2020). Design Thinking Methods to Activate Co-creation Process Among Policymakers, Creative Industries and SMEs. In Proceedings of the II International Triple Helix Summit 2 (pp. 263-277). Springer International Publishing.		x	x	x		x		x	x								x	N/A	-Management support -Ensure funding -Having clear metrics	-Diversity orientation -Foster empathy -Establish collaboration	-Access to user -Provide necessary material -Lessons learnt	-Provide training. -Collaborative initiative key partners -Team skills

Table 1. *Cont.*

			Principles					Criteria				Phases					Levels of Innovation			Tools	Critical Factors			
N	Year	References (in Chronological Order)	Human Centered	Future Oriented	Holistic Systemic	Collaborative	Experimental	Desirability	Feasibility	Viability	Sustainability–Circularity	Empathize	Ideate	Define	Prototype	Test	Product–Technology	Business Models-Value Chain	Ecosystems	Design Thinking Tools	Strategy	Culture	Implementation	Competences
10	2019	[89] Redante, R. C., de Medeiros, J. F., Vidor, G., Cruz, C. M. L., & Ribeiro, J. L. D. (2019). Creative approaches and green product development: Using design thinking to promote stakeholders' engagement. Sustainable Production and Consumption, 19, 247-256.			x	x	x		x								x			N/A	Management support	-Foster empathy -Establish collaboration	-Access to user -Integrate DT into NPD	-Collaborative initiative key partners -DT principles
11	2019	[90] Félix, M. J., Gonçalves, S., Jimenez, G., & Santos, G. (2019). The contribution of design to the development of products and manufacturing processes in the Portuguese industry. Procedia Manufacturing, 41, 1055-1062.		x	x	x	x		x								x			N/A	Management support	Ability to handle Ambiguitious situation	Access to user	Team skills
12	2018	[76] Heck, J., Rittiner, F., Meboldt, M., & Steinert, M. (2018). Promoting user-centricity in short-term ideation workshops. International Journal of Design Creativity and Innovation, 6(3-4), 130-145.	x		x	x	x					x		x		x	x	x		Persona	Having clear metrics	-Diversity orientation -Foster empathy -Establish collaboration	-Access to user -Innovation spaces -Integrate DT into NPD	-Collaborative initiative key partners -DT principles -Team skills
13	2018	[78] O'Gorman, P., Morgan, M., & Van Merkom, R. (2018, December). Using QFD to Normalize a Culture of Innovation in an Engineering SME. In 2018 IEEE International Conference on Industrial Engineering and Engineering Management (IEEM) (pp. 307-310). IEEE.					x		x				x			x	x	x			QFD	Management support	Establish collaboration	-Access to user -Establish flexible processes	-Collaborative initiative key partners
14	2016	[91] Roach, D. C., Ryman, J. A., & Makani, J. (2016). Effectuation, innovation and performance in SMEs: an empirical study. European Journal of Innovation Management, 19(2), 214-238.				x	x				x						x	x		N/A	Having clear metrics	-Ability to handle ambiguitious situation -Ability to handle complexity	Access to user	Collaborative initiative key partners
15	2016	[92] Gerlitz, L.; Hack, A.; Prause, G. (2016). An Integrated Design Management Concept: Creating Innovative Space for Emergent SMEs and Value for Knowledge Absorbers. Journal of Entrepreneurship and Innovation in Emerging Economies, 2(1), 38–55. doi:10.1177/2393957515619717					x		x	x	x						x	x		N/A	-Management support -Having clear metrics	-Diversity orientation -Ability to handle complexity -Establish collaboration	-Access to user -Innovation spaces -Integrate DT into NPD	-Provide training. -Team skills
16	2016	[77] West, S., & Di Nardo, S. (2016). Creating product-service system opportunities for small and medium size firms using service design tools. Procedia CIRP, 47, 96-101.					x				x	x		x		x	x			Empathy map	Having clear metrics	-Foster empathy -Ability to handle complexity	Access to user	Team skills
							x				x	x	x			x	x			Job to be done				
							x				x	x	x	x	x	x	x			Customer side				
							x				x	x	x	x	x	x	x			Supplier side				
17	2015	[93] Gulari, M.N, & Fremantle, C. (2015). Are design-led innovation approaches applicable to SMEs?. In DS 82: Proceedings of the 17th International Conference on Engineering and Product Design Education (E&PDE15), Great Expectations: Design Teaching, Research & Enterprise, Loughborough, UK, 03-04.09. 2015 (pp. 556-561).	x		x	x			x	x							x			N/A	Management support	Ability to handle complexity and uncertainty	Access to user	Collaborative initiative key partners

Table 1. *Cont.*

			Principles					Criteria				Phases					Levels of Innovation			Tools	Critical Factors			
N	Year	References (in Chronological Order)	Human Centered	Future Oriented	Holistic Systemic	Collaborative	Experimental	Desirability	Feasibility	Viability	Sustainability–Circularity	Empathize	Ideate	Define	Prototype	Test	Product–Technology	Business Models-Value Chain	Ecosystems	Design Thinking Tools	Strategy	Culture	Implementation	Competences
18	2023	[94] Rösch, N., Tiberius, V., & Kraus, S. (2023). Design thinking for innovation: context factors, process, and outcomes. European Journal of Innovation Management, 26(7), 160-176.	x		x	x	x	x	x	x	x	x	x	x	x	x	x	x		Participant observation	-Management support -Guidelines for DT	-Foster empathy -Ability to handle ambiguitious situation -Ability to handle complexity -Establish collaboration	-Access to user -Integrate DT into NPD -Lessons learnt	-Collaborative initiative key partners -DT principles -Team skills -Provide training
			x		x	x	x	x	x	x	x	x	x	x	x	x	x	x		Interviewing				
			x		x	x	x	x	x	x	x	x		x	x	x	x	x		Journey mapping				
			x		x	x	x	x	x	x	x	x	x			x	x	x		Job-to-be-done analysis				
19	2022	[95] Manzke, J. (2022). Design Thinking for Innovation Within Manufacturing SMEs: A Multiple Case Study (Doctoral dissertation, Walden University).	x		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	Observation, interviews	-Management support -Having clear metrics	-Foster empathy -Ability to handle complexity and uncertainty -Establish collaboration	-Access to user -Establish flexible processes. -Innovation spaces	-Provide training. -Collaborative initiative key partners -Team skills -DT principles
			x		x	x	x	x	x	x	x	x		x	x	x	x	x	x	Customer journey mapping, co-creation				
			x		x	x	x	x	x	x	x		x		x	x	x	x	x	Brainstorming, co-creation				
			x		x	x	x	x	x	x	x		x		x	x	x	x	x	Experimentation rapid prototyping				
20	2022	[96] Nakata, C., & Bahadir, S. C. (2022). Managing design for innovative new products and services. Journal of Business Strategy, 43(5), 275-282.	x			x	x			x							x			N/A	Management support	Establish collaboration	-Access to user -Lessons learnt	Provide training.
21	2022	[27] Eisenbart, B., Bouwman, S., Voorendt, J., McKillagan, S., Kuys, B., & Ranscombe, C. (2022). Implementing design thinking to drive innovation in technical design. International Journal of Design Creativity and Innovation, 10(3), 141-160.	x	x	x	x	x		x	x	x	x	x	x	x	x	x	x		User research (including interviews and / or focus groups)	Having clear metrics	-Diversity orientation -Foster empathy -Ability to handle complexity and uncertainty -Establish collaboration	-Establish flexible processes. -Access to user -Integrate DT into NPD	-Collaborative initiative key partners -Team skills -DT principles
			x	x	x	x	x		x	x	x	x		x	x	x	x	x		User/customer journey mapping				
			x	x	x	x	x		x	x	x	x		x		x	x	x		Personas				
			x	x	x	x	x		x	x	x	x	x	x	x	x	x	x		Co-creation with users				
			x	x	x	x	x		x	x	x		x		x	x	x	x		User stories				
			x	x	x	x	x		x	x	x	x	x	x	x	x	x	x		Building a point of view.				
22	2022	[97] Moretti, D. M., Baum, C. M., Wustmans, M., & Bröring, S. (2022). Application of journey maps to the development of emergent sustainability-oriented technologies: Lessons for user involvement in agriculture. Business Strategy & Development, 5(3), 209-221.	x		x	x	x	x	x		x	x		x	x	x	x	x		Journey map	Having clear metrics	-Diversity orientation -Ability to handle complexity and uncertainty -Establish collaboration	-Access to user -Integrate DT into NPD	-Collaborative initiative key partners -Team skills

Table 1. *Cont.*

			Principles					Criteria				Phases					Levels of Innovation			Tools	Critical Factors			
N	Year	References (in Chronological Order)	Human Centered	Future Oriented	Holistic Systemic	Collaborative	Experimental	Desirability	Feasibility	Viability	Sustainability–Circularity	Empathize	Ideate	Define	Prototype	Test	Product–Technology	Business Models-Value Chain	Ecosystems	Design Thinking Tools	Strategy	Culture	Implementation	Competences
23	2021	[98] Starostka, J., Evald, M. R., Clarke, A. H., & Hansen, P. R. (2021). Taxonomy of design thinking facilitation. Creativity and Innovation Management, 30(4), 836-844.	x			x	x	x									x	x		N/A	Having clear metrics	-Foster empathy -Establish collaboration	-Access to user	-Collaborative initiative key partners -Team skills -DT principles
24	2020	[99] Nakata, C. (2020). Design thinking for innovation: Considering distinctions, fit, and use in firms. Business horizons, 63(6), 763-772.	x			x	x		x	x							x	x		N/A	Having clear metrics	-Foster empathy -Ability to handle complexity and uncertainty -Establish collaboration	-Access to user	-Collaborative initiative key partners -Team skills
25	2020	[100] Wrigley, C., Nusem, E., & Straker, K. (2020). Implementing design thinking: Understanding organizational conditions. California Management Review, 62(2), 125-143.	x			x		x	x	x							x	x		N/A	-Management support -Having clear metrics	-Foster empathy -Diversity orientation -Establish collaboration	-Access to user -Integrate DT into NPD -Innovation spaces	-Provide training -Collaborative initiative key partners -DT awareness -Team skills -DT principles
26	2020	[101] Savić, M. (2020). Key resources in small and medium enterprises for business model innovation.	x			x	x		x	x	x						x	x	x	N/A	Management support	-Foster empathy -Establish collaboration	Access to user	-Provide training -Collaborative initiative key partners
27	2019	[29] De Paula, D., Dobrigkeit, F., & Cormican, K. (2019, July). Doing it right-critical success factors for design thinking implementation. In Proceedings of the Design Society: International Conference on Engineering Design (Vol. 1, No. 1, pp. 3851-3860). Cambridge University Press.	x		x	x	x				x						x			N/A	-Management support -Guidelines for DT -Funding for DT -Having clear metrics	-Diversity orientation -Foster empathy -Ability to handle ambiguitious situation -Ability to handle complexity and uncertainty -Establish collaboration	-Provide material -Innovation spaces -Establish flexible processes. -Integrate DT into NPD -Lessons learnt -Access to user	-Provide training -Collaborative initiative key partners -DT awareness -Team skills -DT principles

Table 1. *Cont.*

			Principles					Criteria				Phases					Levels of Innovation			Tools	Critical Factors			
N	Year	References (in Chronological Order)	Human Centered	Future Oriented	Holistic Systemic	Collaborative	Experimental	Desirability	Feasibility	Viability	Sustainability–Circularity	Empathize	Ideate	Define	Prototype	Test	Product–Technology	Business Models-Value Chain	Ecosystems	Design Thinking Tools	Strategy	Culture	Implementation	Competences
28	2018	[102] Lugnet, J., Wenngren, J., & Ericson, Å. (2018). ADDRESSING TEAM BASED INNOVATION FOR SMALL FIRMS-CREATE, BUILD, TEST & LEARN. In DS 92: Proceedings of the DESIGN 2018 15th International Design Conference (pp. 849-856).				x	x										x			N/A	Having clear metrics	-Foster empathy -Ability to handle ambiguitious situation -Ability to handle complexity and uncertainty -Establish collaboration	-Lessons learnt -Access to user	-Provide training -Team skills -DT principles
29	2016	[32] Carlgren, L., Elmquist, M., & Rauth, I. (2016). The challenges of using design thinking in industry –experiences from five large firms. Creativity and Innovation Management, 25(3), 344-362.	x		x	x	x		x	x	x						x	x		N/A	Having clear metrics	-Diversity orientation -Foster empathy -Ability to handle complexity and uncertainty -Establish collaboration	-Integrate DT into NPD -Lessons learnt -Access to user	-Provide training -Collaborative initiative key partners -DT awareness -Team skills -DT principles
30	2011	[103] Malins, J. P. (2011). Innovation by design: using design thinking to support SMEs.				x					x						x	x		N/A	Funding for DT	-Foster empathy -Ability to handle ambiguitious situation -Establish collaboration	-Innovation spaces -Lessons learnt -Access to user	DT principles

References

1. Bonakdar, A.; Gassmann, O. Design Thinking for Revolutionizing Your Business Models. In *Design Thinking for Innovation: Research and Practice*; Springer: Berlin/Heidelberg, Germany, 2016; pp. 57–66.
2. Martin, B.; Hanington, B. *Universal Methods of Design: 100 WAYS to Research Complex Problems, Develop Innovative Ideas, and Design Effective Solutions*; Rockport: Beverly, MA, USA, 2012.
3. Klewitz, J.; Hansen, E.G. Sustainability-oriented innovation of SMEs: A systematic review. *J. Clean. Prod.* **2014**, *65*, 57–75. [CrossRef]
4. De Jong, J.P.J.; Marsili, O. The fruit flies of innovations: A taxonomy of innovative small firms. *Res. Policy* **2006**, *35*, 213–229. [CrossRef]
5. Kenney, M. The Temporal Dynamics of Knowledge Creation in the Information Society. In *Knowledge Emergence, Social, Technical, and Evolutionary Dimensions of Knowledge Creation*; Oxford University Press: New York, NY, USA, 2001.
6. Andreassen, T.W.; Lervik-Olsen, L.; Calabretta, G. Deriving Valuable Innovations. Ph.D. Thesis, Norwegian School of Economics, Bergen, Norway, 2015.
7. Geissdoerfer, M.; Bocken, N.M.P.; Hultink, E.J. Design thinking to enhance the sustainable business modelling process e A workshop based on a value mapping process. *J. Clean. Prod.* **2016**, *135*, 1218–1232. [CrossRef]
8. Iqbal, M.; Suzianti, A. New product development process design for small and medium enterprises: A systematic literature review from the perspective of open innovation. *J. Open Innov. Technol. Mark. Complex.* **2021**, *7*, 153. [CrossRef]
9. Leithold, N.; Haase, H.; Lautenschläger, A. Stage-Gate® for SMEs: A qualitative study in Germany. *Eur. J. Innov. Manag.* **2015**, *18*, 130–149. [CrossRef]
10. Elsbach, K.D.; Stigliani, I. Design Thinking and Organizational Culture: A Review and Framework for Future Research. *J. Manag.* **2018**, *44*, 2274–2306. [CrossRef]
11. Micheli, P.; Perks, H.; Beverland, M.B. Elevating Design in the Organization. *J. Prod. Innov. Manag.* **2018**, *35*, 629–651. [CrossRef]
12. Cousins, B. Design thinking: Organizational learning in VUCA environments. *Acad. Strateg. Manag. J.* **2018**, *17*, 1–18.
13. Ferrara, M.; Di Milano, P.; Chiara Lecce, I. "Design for Enterprises": Developing European SMEs capabilities for design-driven innovation. *Mark. Glob. Dev. Rev.* **2020**, *4*, 1–27. [CrossRef]
14. Brown, T. Design Thinking. *Harv. Bus. Rev.* **2008**, *86*, 84–92. [PubMed]
15. Liedtka, J. Perspective: Linking Design Thinking with Innovation Outcomes through Cognitive Bias Reduction. *J. Prod. Innov. Manag.* **2015**, *32*, 925–938. [CrossRef]
16. Micheli, P.; Wilner, S.J.S.; Bhatti, S.H.; Mura, M.; Beverland, M.B. Doing Design Thinking: Conceptual Review, Synthesis, and Research Agenda. *J. Prod. Innov. Manag.* **2019**, *36*, 124–148. [CrossRef]
17. Kelley, T.; Kelley, D. *Creative Confidence: Unleashing the Creative Potential Within Us All*; Crown Business: New York, NY, USA, 2013.
18. Liedtka, J.; Ogilvie, T. *Designing for Growth: A Design Thinking Toolkit for Managers*; Columbia Business School Publishing: New York, NY, USA, 2011.
19. Martin, R. The Design of Business: Why Design Thinking Is the next Competitive Advantage-DESIGNING A BUSINESS. 2009.
20. Acklin, C. Design-Driven Innovation Process Model. *Des. Manag. J.* **2010**, *5*, 50–60. [CrossRef]
21. Carlgren, L.; Rauth, I.; Elmquist, M. Framing Design Thinking: The Concept in Idea and Enactment. *Creat. Innov. Manag.* **2016**, *25*, 38–57. [CrossRef]
22. Best, K. Design Management: Managing Design Strategy. In *Process and Implementation*; AVA Publishing: Brooklyn, NY, USA, 2006.
23. Wölbling, A.; Krämer, K.; Buss, C.N.; Dribbisch, K.; LoBue, P.; Taherivand, A. Design Thinking: An Innovative Concept for Developing User-Centered Software. In *Software for People*; Management for Professionals; Springer: Berlin/Heidelberg, Germany, 2012; Part F393; pp. 121–136.
24. Brenner, W.; Uebernickel, F.; Abrell, T. Design Thinking as Mindset, Process, and Toolbox. In *Design Thinking for Innovation: Research and Practice*; Springer: Berlin/Heidelberg, Germany, 2016; pp. 3–21.
25. Stickdorn, M.; Schneider, J.; Andrews, K.; Lawrence, A. *This Is Service Design Thinking: Basics, Tools, Cases*; Wiley: Hoboken, NJ, USA, 2011; p. 1.
26. Hassi, L.; Laakso, M. Design thinking in the management discourse: Defining the elements of the concept. In Proceedings of the 18th International Product Development Management Conference, IPDMC 2011, Delft, The Netherlands, 5–7 June 2011; pp. 1–14.
27. Eisenbart, B.; Bouwman, S.; Voorendt, J.; McKillagan, S.; Kuys, B.; Ranscombe, C. Implementing design thinking to drive innovation in technical design. *Int. J. Des. Creat. Innov.* **2022**, *10*, 141–160. [CrossRef]
28. Eybers, S. Identifying Critical Success Factors for Business Intelligence Systems. In Proceedings of the European Conference on Information Systems Management, Academic Conferences International Limited, Münster, Germany, 26–29 May 2015; p. 77.
29. de Paula, D.; Dobrigkeit, F.; Cormican, K. Doing it right-Critical success factors for design thinking implementation. In Proceedings of the International Conference on Engineering Design (ICED), Delft, The Netherlands, 5–8 August 2019; Cambridge University Press: Cambridge, UK, 2019; pp. 3851–3860.
30. Holloway, M. How tangible is your strategy? How design thinking can turn your strategy into reality. *J. Bus. Strategy* **2009**, *30*, 50–56. [CrossRef]
31. Rosensweig, R.R. More than Heroics: Building Design as a Dynamic Capability. *Des. Manag. J.* **2011**, *6*, 16–26. [CrossRef]
32. Carlgren, L.; Elmquist, M.; Rauth, I. The Challenges of Using Design Thinking in Industry–Experiences from Five Large Firms. *Creat. Innov. Manag.* **2016**, *25*, 344–362. [CrossRef]

33. Kimbell, L. Beyond design thinking: Design-as-practice and designs-in-practice. In Proceedings of the CRESC Conference, Manchester, UK, 1–4 September 2009.
34. Liedtka, J. Learning to use design thinking tools for successful innovation. *Strategy Leadersh.* **2011**, *39*, 13–19. [CrossRef]
35. Lockwood, T. Transition: How to Become a More Design-Minded Organization. *Des. Manag. Rev.* **2009**, *20*, 28–37. [CrossRef]
36. Dunne, D.; Martin, R.; Rotman, J.L. Design Thinking and How It Will Change Management Education: An Interview and Discussion. *Acad. Manag. Learn. Educ.* **2006**, *5*, 512–523. [CrossRef]
37. Gloppen, J. Perspectives on Design Leadership and Design Thinking and How They Relate to European Service Industries. *Des. Manag. J. Des. Manag. J.* **2009**, *4*, 33–47. [CrossRef]
38. Sato, S.; Lucente, S.; Mrazek, D. Design thinking to make organization change and development more responsive. *Des. Manag. Rev.* **2010**, *21*, 44–52. [CrossRef]
39. Boland, R.J.; Collopy, F. Design matters for management. In *Managing as Designing*; Stanford University Press: Redwood City, CA, USA, 2004; pp. 3–18.
40. Cooper, R.; Junginger, S.; Lockwood, T. Design Thinking and Design Management: A Research and Practice Perspective. *Des. Manag. Rev.* **2009**, *20*, 46–55. [CrossRef]
41. Dew, N. Abduction: A pre-condition for the intelligent design of strategy. *J. Bus. Strategy* **2007**, *28*, 38–45. [CrossRef]
42. Micheli, P.; Jaina, J.; Goffin, K.; Lemke, F.; Verganti, R. Perceptions of industrial design: The "means" and the "ends. *" J. Prod. Innov. Manag.* **2012**, *29*, 687–704. [CrossRef]
43. Carlgren, L.; Elmquist, M.; Rauth, I. Design Thinking: Exploring Values and Effects from an Innovation Capability Perspective. *Des. J.* **2014**, *17*, 403–423. [CrossRef]
44. Ward, A.; Runcie, E.; Morris, L. Embedding innovation: Design thinking for small enterprises. *J. Bus. Strategy* **2009**, *30*, 78–84. [CrossRef]
45. Seidel, V.P.; Fixson, S.K. Adopting Design Thinking in Novice Multidisciplinary Teams: The Application and Limits of Design Methods and Reflexive Practices. *J. Prod. Innov. Manag.* **2013**, *30*, 19–33. [CrossRef]
46. Donthu, N.; Kumar, S.; Mukherjee, D.; Pandey, N.; Lim, W.M. How to conduct a bibliometric analysis: An overview and guidelines. *J. Bus. Res.* **2021**, *133*, 285–296. [CrossRef]
47. Passas, I. Bibliometric Analysis: The Main Steps. *Encyclopedia* **2024**, *4*, 1014–1025. [CrossRef]
48. Klarin, A. How to conduct a bibliometric content analysis: Guidelines and contributions of content co-occurrence or co-word literature reviews. *Int. J. Consum. Stud.* **2024**, *48*, e13031. [CrossRef]
49. Börner, K.; Chen, C.; Boyack, K.W. Visualizing Knowledge Domains. *Annu. Rev. Inf. Sci. Technol.* **2003**, *37*, 179–255. [CrossRef]
50. Chen, X.; Zou, D.; Xie, H. Fifty years of British Journal of Educational Technology: A topic modeling based bibliometric perspective. *Br. J. Educ. Technol.* **2020**, *51*, 692–708. [CrossRef]
51. van Eck, N.J.; Waltman, L. Visualizing Bibliometric Networks. In *Measuring Scholarly Impact*; Springer International Publishing: Cham, Switzerland, 2014; pp. 285–320.
52. Baier-Fuentes, H.; Cascón-Katchadourian, J.; Sánchez, Á.M.; Herrera Viedma, E. A Bibliometric Overview of the International Journal of Interactive Multimedia and Artificial Intelligence. *Int. J. Interact. Multimed. Artif. Intell.* **2018**, *5*, 9. [CrossRef]
53. Melazzini, M.; Campodall'Orto, S.; Carella, G.; Vignati, A.; Zurlo, F. Design thinking methods to activate co-creation process among policymakers, creative industries and SMEs. *Lect. Notes Civ. Eng.* **2020**, *43*, 263–277.
54. Konietzko, J.; Bocken, N.; Hultink, E.J. A tool to analyze, ideate and develop circular innovation ecosystems. *Sustainability* **2020**, *12*, 417. [CrossRef]
55. Rashid, A.; Asif, F.M.A.; Krajnik, P.; Nicolescu, C.M. Resource conservative manufacturing: An essential change in business and technology paradigm for sustainable manufacturing. *J. Clean. Prod.* **2013**, *57*, 166–177. [CrossRef]
56. Diehl, J.C.; Christiaans, H. Product Service Systems: The Future for Designers? The Changing Role of the Industrial Designer. In Proceedings of the International Design Congress of KSDS and Adada with Cumulus, Gwangju, Republic of Korea, 17–23 October 2015; pp. 17–23.
57. Chesbrough, H. Business Model Innovation: Opportunities and Barriers. *Long. Range Plann.* **2010**, *43*, 354–363. [CrossRef]
58. Tukker, A. Eight types of product–service system: Eight ways to sustainability? Experiences from SusProNet. *Bus. Strategy Environ.* **2004**, *13*, 246–260. [CrossRef]
59. Prendeville, S.; Cherim, E.; Bocken, N. Circular Cities: Mapping Six Cities in Transition. *Environ. Innov. Soc. Transit.* **2018**, *26*, 171–194. [CrossRef]
60. Guldmann, E.; Bocken, N.M.P.; Brezet, H. A Design Thinking Framework for Circular Business Model Innovation. *J. Bus. Models* **2019**, *7*, 39–70.
61. Calabretta, G.; Gemser, G.; Karpen, I. *Strategic Design: Eight Essential Practices Every Strategic Designer Must Master*; BIS Publishers: Amsterdam, The Netherlands, 2016.
62. Baldassarre, B.; Konietzko, J.; Brown, P.; Calabretta, G.; Bocken, N.; Karpen, I.O.; Hultink, E.J. Addressing the design-implementation gap of sustainable business models by prototyping: A tool for planning and executing small-scale pilots. *J. Clean. Prod.* **2020**, *255*, 120295. [CrossRef]
63. Bocken, N.; Baldassarre, B.; Keskin, D.; Diehl, J.C. Design thinking tools to catalyse sustainable circular innovation. In *The Routledge Handbook of Catalysts for a Sustainable Circular Economy*; Routledge: London, UK, 2023; pp. 1–36.

64. Henriksen, D.; Richardson, C.; Mehta, R. Design thinking: A creative approach to educational problems of practice. *Think. Ski. Creat.* **2017**, *26*, 140–153. [CrossRef]
65. Kwon, J.; Choi, Y.; Hwang, Y. Enterprise Design Thinking: An Investigation on User-Centered Design Processes in Large Corporations. *Designs* **2021**, *5*, 43. [CrossRef]
66. Design Council. Framework for Innovation: Design Council's Evolved Double Diamond. Available online: https://www.designcouncil.org.uk/our-resources/framework-for-innovation/ (accessed on 11 December 2022).
67. Dorst, K. The core of "design thinking" and its application. *Des. Stud.* **2011**, *32*, 521–532. [CrossRef]
68. Pap, M.; Vdović, R.; Baletić, B. Design Thinking metoda u znanstvenom istraživanju, edukaciji i poslovnoj praksi. *Prostor* **2019**, *27*, 334–347. [CrossRef]
69. Antoljak, V.; Kosović, M. *Design Thinking za Nedizajnere–Kako Riješiti Poslovne Probleme i Uspješno Inovirati*; Školska Knjiga: Zagreb, Croatia, 2018. (In Croatian)
70. Starzyńska, B.; Kujawińska, A.; Górski, F.; Buń, P. Virtual quality toolbox as an innovative solution supporting lifelong learning. *Int. J. Qual. Res.* **2022**, *16*, 1013–1028. [CrossRef]
71. Lemon, K.N.; Verhoef, P.C. Understanding customer experience throughout the customer journey. *J. Mark.* **2016**, *80*, 69–96. [CrossRef]
72. Luka, I. Design Thinking in Pedagogy. *J. Educ. Cult. Soc.* **2014**, *5*, 63–74. [CrossRef]
73. Wolfe, R.A.; Putler, D.S. How Tight Are the Ties that Bind Stakeholder Groups? *Organ. Sci.* **2002**, *13*, 64–80. [CrossRef]
74. Bettencourt, L.A.; Ulwick, A.W. The Customer-Centered Innovation Map. *Harv. Bus. Rev.* **2008**, *86*, 109. [PubMed]
75. Long, F. Real or imaginary: The effectiveness of using personas in product design. In Proceedings of the Irish Ergonomics Society Annual Conference, Dublin, Ireland, 14 May 2009; Volume 14, pp. 1–10.
76. Heck, J.; Rittiner, F.; Meboldt, M.; Steinert, M. Promoting user-centricity in short-term ideation workshops. *Int. J. Des. Creat. Innov.* **2018**, *6*, 130–145. [CrossRef]
77. West, S.; Di Nardo, S. Creating Product-service System Opportunities for Small and Medium Size Firms Using Service Design Tools. *Procedia CIRP* **2016**, *47*, 96–101. [CrossRef]
78. O'Gorman, P.; Morgan, M.; Van Merkom, R. Using QFD to Normalize a Culture of Innovation in an Engineering SME. In Proceedings of the 2018 IEEE International Conference on Industrial Engineering and Engineering Management, Bangkok, Thailand, 16–19 December 2018.
79. Rittershaus, P.; Renner, M.; Aryan, V. A conceptual methodology to screen and adopt circular business models in small and medium scale enterprises (SMEs): A case study on child safety seats as a product service system. *J. Clean. Prod.* **2023**, *390*, 136083. [CrossRef]
80. Glen, R.; Suciu, C.; Baughn, C.C.; Anson, R. Teaching design thinking in business schools. *Int. J. Manag. Educ.* **2015**, *13*, 182–192. [CrossRef]
81. Garbuio, M.; Dong, A.; Lin, N.; Tschang, T.; Lovallo, D. Demystifying the Genius of Entrepreneurship: How Design Cognition Can Help Create the Next Generation of Entrepreneurs. *Acad. Manag. Learn. Educ.* **2018**, *17*, 41–61. [CrossRef]
82. Björk, J.; Karlsson, M.P.; Magnusson, M. Turning ideas into innovations-introducing demand-driven collaborative ideation. *Int. J. Innov. Reg. Dev.* **2014**, *5*, 429. [CrossRef]
83. Paay, J.; Kuys, B.; Taffe, S. Innovating product design through university-industry collaboration: Codesigning a bushfire rated skylight. *Des. Stud.* **2021**, *76*, 101031. [CrossRef]
84. Gao, B.; Yu, K. Knowledge exchange in SMEs service innovation with design thinking. *Manag. Decis.* **2023**, *61*, 2029–2049. [CrossRef]
85. Baumkötter, B.; Ellenbürger, J.; Glemser, M.; Raschi, M. Neue Produkte erfordern neue Denkweisen zur Einführung: Entwicklung eines neuartigen Markteinführungskonzeptes (in German). *ZWF Zeitschrift fuer Wirtschaftlichen Fabrikbetrieb* **2023**, *118*, 232–236. [CrossRef]
86. Storm, J.; Smith, A. Empathize with Whom? Adopting a Design Thinking Mind-Set to Stimulate Sustainability Initiatives in Chinese SMEs. *Sustainability* **2023**, *15*, 252. [CrossRef]
87. Whelan, L.; Kiernan, L.; Morrissey, K.; Deloughry, N. Measuring the success factors of strategic design implementation. In Proceedings of the 2021 IEEE Technology and Engineering Management Conference-Europe, TEMSCON-EUR 2021, Dubrovnik, Croatia, 17–20 May 2021.
88. Lugnet, J.; Ericson, Å.; Wenngren, J. Innovation Supports for Small-Scale Development in Rural Regions: A Create, Build, Test and Learn Approach. *Int. J. Prod. Development.* **2020**, *24*, 30. [CrossRef]
89. Redante, R.C.; de Medeiros, J.F.; Vidor, G.; Lima Cruz, C.M. Creative approaches and green product development: Using design thinking to promote stakeholders' engagement. Sustain. *Prod. Consum.* **2019**, *19*, 247–256. [CrossRef]
90. Félix, M.J.; Gonçalves, S.; Jimenez, G.; Santos, G. The Contribution of Design to the Development of Products and Manufacturing Processes in the Portuguese Industry. *Procedia. Manuf.* **2019**, *41*, 1055–1062. [CrossRef]
91. Roach, D.C.; Ryman, J.A.; Makani, J. Effectuation, innovation and performance in SMEs: An empirical study. *Eur. J. Innov. Manag.* **2016**, *19*, 214–238. [CrossRef]
92. Gerlitz, L.; Hack, A.; Prause, G. An Integrated Design Management Concept: Creating Innovative Space for Emergent SMEs and Value for Knowledge Absorbers. *J. Entrep. Innov. Emerg. Econ.* **2016**, *2*, 38–55. [CrossRef]

93. Gulari, M.N.; Fremantle, C. Are Design-Led Innovation Approaches Applicable To SMEs? In Proceeding of International conference on engineering and product design education, Loughborough, UK, 3–4 September 2015.
94. Rösch, N.; Tiberius, V.; Kraus, S. Design thinking for innovation: Context factors, process, and outcomes. *Eur. J. Innov. Manag.* **2023**, *26*, 160–176. [CrossRef]
95. Manzke, J. Design Thinking for Innovation Within Manufacturing SMEs: A Multiple Case Study. *Walden Dissertations and Doctoral Studies*. 2022, p. 12450. Available online: https://scholarworks.waldenu.edu/dissertations/12450 (accessed on 11 December 2022).
96. Nakata, C.; Bahadir, S.C. Managing design for innovative new products and services. *J. Bus. Strategy* **2022**, *43*, 275–282. [CrossRef]
97. Moretti, D.M.; Baum, C.M.; Wustmans, M.; Bröring, S. Application of journey maps to the development of emergent sustainability-oriented technologies: Lessons for user involvement in agriculture. *Bus. Strategy Dev.* **2022**, *5*, 209–221. [CrossRef]
98. Starostka, J.; Evald, M.R.; Clarke, A.H.; Hansen, P.R. Taxonomy of design thinking facilitation. *Creat. Innov. Manag.* **2021**, *30*, 836–844. [CrossRef]
99. Nakata, C. Design thinking for innovation: Considering distinctions, fit, and use in firms. *Bus. Horiz.* **2020**, *63*, 763–772. [CrossRef]
100. Wrigley, C.; Nusem, E.; Straker, K. Implementing Design Thinking: Understanding Organizational Conditions. *Calif Manag. Rev* **2020**, *62*, 125–143. [CrossRef]
101. Savić, M. Key resources in small and medium enterprises for business model innovation. In Proceedings of the 20th International Conference Sustainable Innovation 2015: 'State of the Art' Sustainable Innovation & Design, Epsom Surrey, UK, 9–10 November 2015.
102. Lugnet, J.; Wenngren, J.; Ericson, A. Addressing team based innovation for small firms—create, build, test & learn. In DS 92: Proceedings of the DESIGN 2018 15th International Design Conference, Dubrovnik, Croatia, 21–24 May 2018; Volume 2, pp. 849–856. [CrossRef]
103. Malins, J.P. Innovation by Design: Using Design Thinking to Support SMEs. *Swed. Des. J.* **2011**, *2*, 25–31.

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.